

The Day Repeats Implementation Report

This pass turns the Day Repeats stage into a concrete generation-and-evidence chain for mobile learners: choose one token, append it, run again, manage temporary context, retrieve evidence, ground claims, and watch for fluent unsupported output.

Build passed. No generated PNG assets added.

What Changed

- Upgraded Autoregression and Context Window to richer lesson architecture.
- Kept Journey order unchanged: Autoregression, Context Window, RAG, Grounding, Hallucinations.
- Added five specific lesson interactions for the stage.
- Reworked coded visuals for Autoregression, Context Window, Grounding, and Hallucinations.
- Preserved games, progress rules, checkpoint randomization, badge logic, dependencies, and Glossary Dojo work.
- Added the **Response-so-far** glossary entry.
- Bumped the visible app version to **v0.22.1**.
- Captured 320px and 390px screenshots with zero horizontal overflow.

Stage Chain

sampling chooses one token -> token is appended -> model runs again -> response-so-far grows -> context window limits what remains visible -> RAG can add retrieved evidence -> grounding connects claims to evidence -> hallucinations happen when fluent output outruns support.

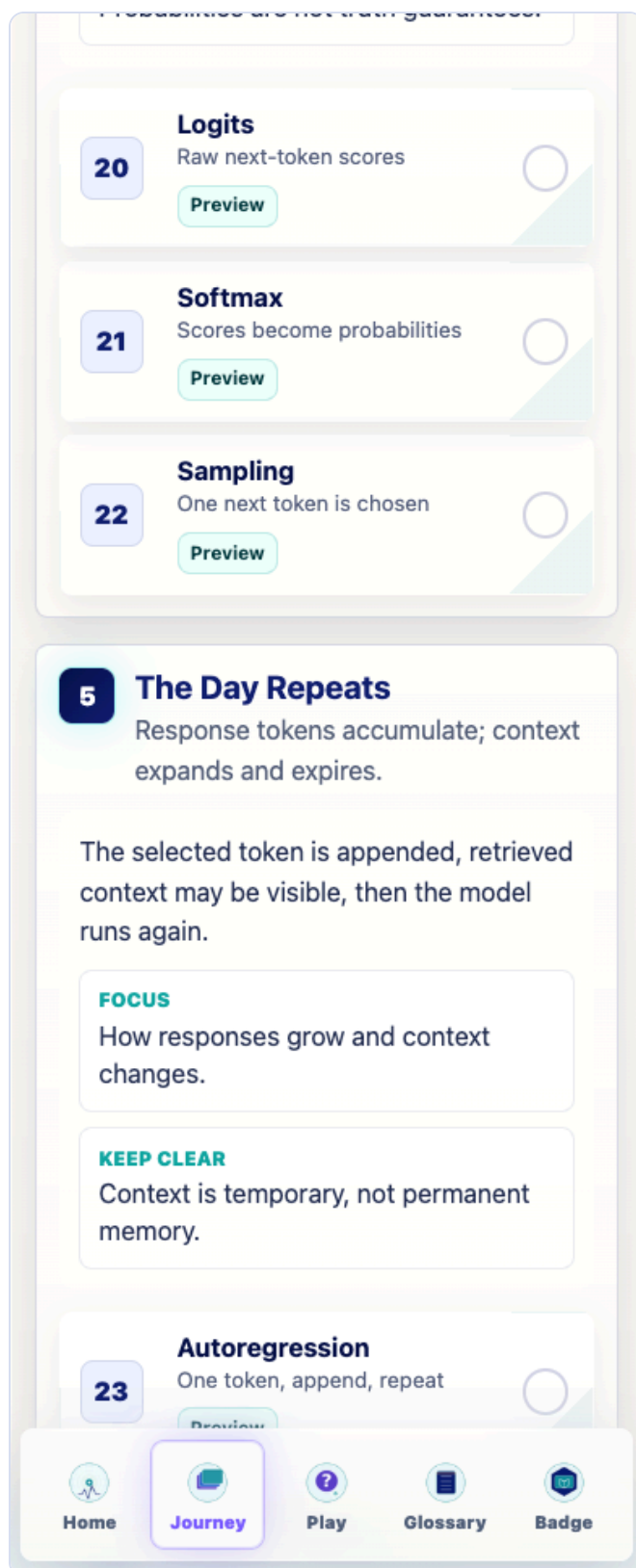
Card Updates

CARD	NEW EMPHASIS	INTERACTION
Autoregression	Uses the pet-conflict example, chosen token <code>floor</code> , appended response-so-far, and repeated forward pass.	<code>autoregression-loop</code>
Context Window	Shows system, user, retrieved note, response-so-far, and old message as temporary context cards.	<code>context-window-tray</code>
RAG and Retrieval	Preserves open-book AI and adds lane highlighting from prompt to response.	<code>rag-lane-highlight</code>
Grounding	Makes claim-to-evidence support visible and warns that citation-looking output is not automatically grounded.	<code>grounding-claim-match</code>
Hallucinations	Frames hallucinations as unsupported/fabricated generation, not lying or intent.	<code>hallucination-support-check</code>

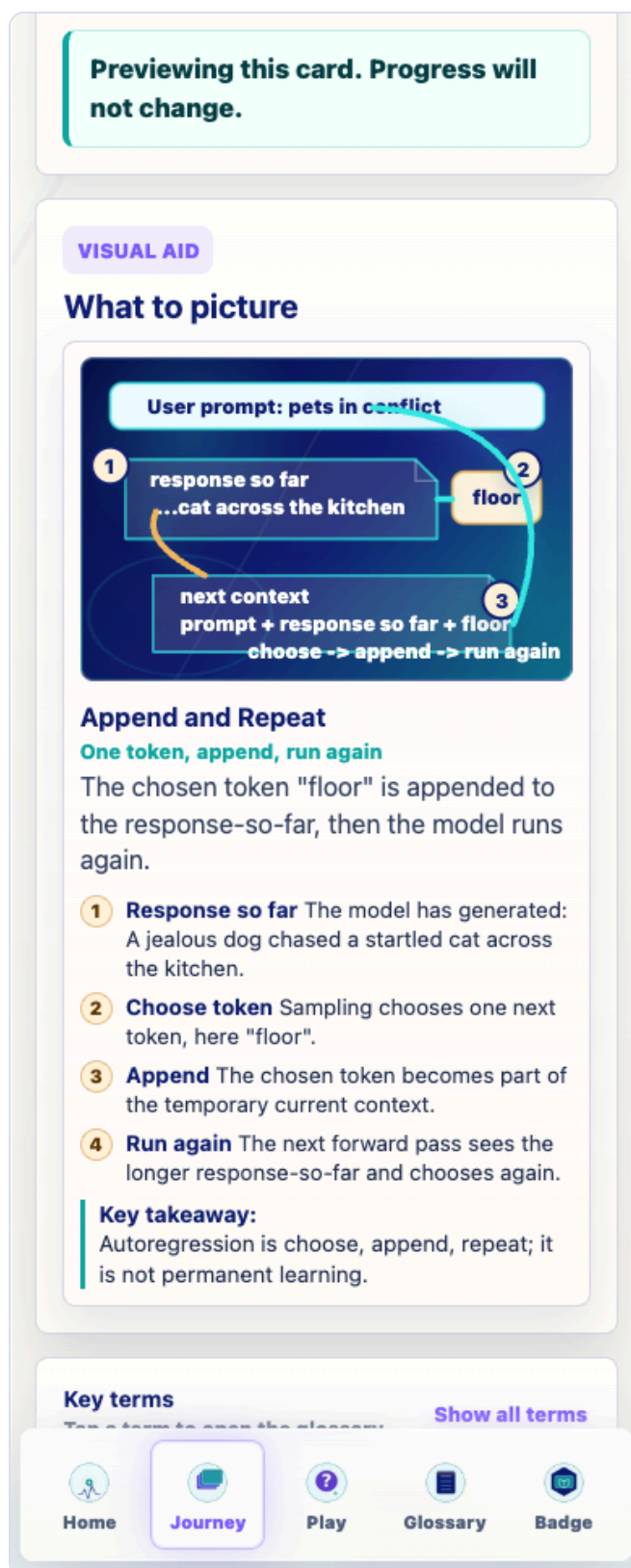
Verification

- `npm run typecheck` : passed.
- `npm run build` : passed with the existing Vite large-chunk warning.
- `npm run build:pages` : passed with the existing Vite large-chunk warning.
- `npm run audit:answers` : passed.
- Screenshot QA: 0 horizontal overflow at 320px and 390px.
- Preview-mode screenshot QA: localStorage progress remained `[]` .

Screenshots



Journey overview keeps the Day Repeats order in place.



Autoregression now shows response-so-far, chosen token, next context, and repeat loop.

TINY INTERACTION

Tap-step append loop

1. So far

2. Token chosen

3. Appended

4. Run again

USER PROMPT

Give me one sentence with two household pets in conflict.

RESPONSE-SO-FAR

A jealous dog chased a startled cat across the kitchen floor

chosen token: floor

. while before

Reset loop

Insight strengthened.
Autoregression means the response grows by choose, append, and repeat.

Tap through choose token, append, and run again to see how the response-so-far grows.

CHECKPOINT



Home



Journey



Play



Glossary

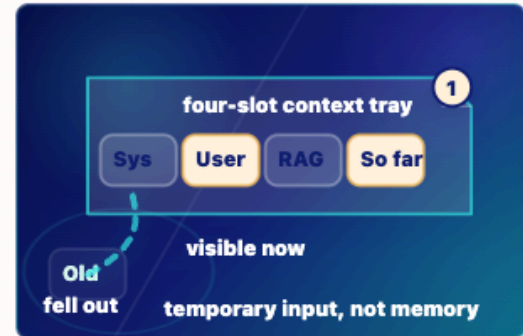


Badge

Previewing this card. Progress will not change.

VISUAL AID

What to picture



Temporary Context Window

Limited visible input

Only the cards still inside the context tray can influence the next token.

- 1 Visible now** System instructions, the user prompt, retrieved notes, and response-so-far may all be current context.
- 2 Limited tray** When the tray fills, older material can fall out or need summarizing by the surrounding system.
- 3 Temporary** Context can feel like memory, but it does not normally update weights or create permanent memory.
- 4 Next token** The model can use only what remains visible in the current run.

Key takeaway:

Context is temporary visible input, not durable memory.

Key terms

[Show all terms](#)



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Journey



Play



Glossary



Badge

Tap-step loop ends with choose, append, and repeat feedback.

Context Window shows a four-slot temporary tray and an old card outside it.

TINY INTERACTION

Push the context cards

Four-slot context window

System

User

RAG note

So far

Fell out

Old message

Reset tray

Insight strengthened.
The model can only use
what remains visible in
the current context.

Tap cards into a four-slot window and watch the oldest card fall out.

CHECKPOINT

What happens to information outside the context window?

The model cannot



Home



Journey



Play



Glossary



Badge

A human can evaluate sources, remember what was read, and reason about trust. The model uses retrieved text as context but does not become aware of it.

TINY INTERACTION

Trace the retrieval lanes

1 Prompt

2 Retriever

3 Notes

4 Context tray

5 Response

Response: The LLM generates response tokens using fixed weights plus current context.

Insight strengthened. RAG adds temporary evidence to context; it does not rewrite model weights.

Tap prompt, retriever, notes, context tray, and response to see retrieval plus context, not training.

CHECKPOINT



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Journey



Play



Glossary



Badge

RAG interaction highlights retrieval plus context, not weight updates.

The 320px tray remains readable after five cards are pushed.

WHERE IT BREAKS

The model is not consciously verifying truth unless the system is designed to check evidence.

TINY INTERACTION

Match claims to evidence

The policy allows X.

Enrollment is 42.

The service is always free.

The service is always free.

NOT SUPPORTED BY

No matching evidence

Not quite. A citation-looking answer is not automatically grounded; the evidence must actually support the claim.

Connect supported claims to evidence and leave unsupported claims unanchored.

CHECKPOINT

What does grounding try to do?

A Make the model conscious



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Journey



Play



Glossary



Badge

A human may intend, deceive, or misremember. The model is generating probable text without human memory or intention.

TINY INTERACTION

Mark missing supports

Which fluent claim needs evidence before you trust it?

The retrieved policy mentions X.
Evidence pillar present

The policy was updated in 2027.
No matching evidence

Exceptions may exist.
Needs review, but not the missing-support claim here

Insight strengthened. A fluent claim can still need grounding or review.

Read three fluent claims and mark the ones that need evidence.

CHECKPOINT

Why can hallucinations happen?

A RAG permanently teaches the model



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Journey



Play



Glossary



Badge

Grounding shows the difference between a claim and matching evidence.

Hallucinations asks learners to mark a fluent claim with missing support.

claim can still need grounding or review.

Read three fluent claims and mark the ones that need evidence.

CHECKPOINT

Why can hallucinations happen?

- A** RAG permanently teaches the model
- B** Softmax guarantees truth
- C** The model can generate plausible text without enough grounding
- D** The model is always trying to deceive

Insight unlocked. Insight strengthened.
A fluent claim can still need grounding or review.

REFLECT

Teach it back in one sentence.

Reflection is available in Learn mode.



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Journey



Play



Glossary



Badge

Checkpoint choices remain randomized and feedback reinforces grounding/review.

Known Issues

- The existing Vite large-chunk warning remains.
- Correct checkpoint feedback still includes the app's global **Insight unlocked.** prefix before lesson-specific feedback; this was left unchanged to avoid a global behavior change.
- No generated Image 2 assets were added; Hallucinations remains the only optional future textless bridge candidate.